Passive RF Intelligence at 915 MHz:

A Systematic Taxonomy of Cleartext Telemetry Across CISA Critical Infrastructure Sectors

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Disclosure status. The findings summarized here were reported to CERT/CC under coordinated disclosure in February 2026 (VRF#26-02-HVGDC, VU#318962), concerning unencrypted telemetry broadcast by Neptune R900 water meter endpoints. Neptune Technology Group, Itron Inc., and Landis+Gyr were invited to the coordination case. The case was closed with the determination that the affected products function as designed; no CVE or advisory was issued. The author subsequently submitted a correction to a factual error in the closing statement, which is on record with the case.

Purpose of this revision. This document was originally prepared as supporting evidence for that case. It is republished as a post-coordination technical reference. Revision 2 separated the author's field observations from literature-derived findings, corrected protocol and citation errors, and added a limitations section. Revision 3 adds author-assessed CVSS v3.1 scores for each taxonomy entry (Section VI-D, Table III); no findings or qualitative ratings changed. The closure determination is itself relevant to the paper's thesis: behavior that is insecure by design, rather than by defect, falls outside the traditional vulnerability-remediation pipeline and requires a policy response.

***Abstract*—**The 902–928 MHz unlicensed Industrial, Scientific, and Medical (ISM) band is a dense source of passively receivable telemetry in North America. Using commodity Software-Defined Radio (SDR) hardware and open-source decoders, a receive-only observer can decode plaintext telemetry from utility meters and consumer IoT sensors without transmitting. Published protocol documentation indicates similar exposure for some industrial telemetry radios and building automation systems. This paper presents a literature-grounded taxonomy of 915 MHz cleartext protocols mapped to the 16 critical infrastructure sectors defined by CISA under PPD-21 and reaffirmed by the April 2024 National Security Memorandum on Critical Infrastructure Security and Resilience. For each device class we document the protocol and modulation, the plaintext fields exposed, the privacy and operational implications established by prior research, a qualitative risk rating, and an author-assessed CVSS v3.1 score. Each entry is labeled as either observed in the author's own receive-only captures or documented from public sources. The author's captures at a single residential location decoded Neptune R900, Itron ERT-SCM and SCMplus, Landis+Gyr GS-series, and Fine Offset WH51 transmissions. Prior work has shown that smart meter consumption data supports high-accuracy occupancy inference; we show that the R900's broadcast status fields, including days of no flow over a 35-day window, expose absence-related information without any interval analysis. Under CVSS v3.1 these passive exposures score no higher than 6.5 (Medium), a ceiling that understates their aggregate privacy impact. Because no federal rule requires encryption for unlicensed 902–928 MHz transmissions and deployed AMR endpoints are typically not field-upgradable, this exposure is likely to persist for the service life of installed hardware absent regulatory action.

Index Terms—902–928 MHz ISM band, Automatic Meter Reading, Software-Defined Radio, critical infrastructure security, ERT protocol, Neptune R900, SCADA wireless, passive surveillance, CISA, CVSS, occupancy detection.

I. Introduction

The 902–928 MHz ISM band, governed by FCC Part 15 (47 CFR Part 15), is license-free spectrum shared by a large installed base of endpoints spanning electric and water utilities, industrial telemetry, building automation, and agricultural monitoring. Part 15 devices operate on a non-interference basis. No FCC rule, NERC CIP standard, or EPA drinking water regulation requires cryptographic protection for over-the-air transmissions in this band [1].

The Encoder Receiver Transmitter (ERT) protocol family, developed by Itron and described in U.S. Patent 4,614,945, underlies much of the Automatic Meter Reading (AMR) infrastructure in North America. ERT transmits its payload without encryption using Manchester-coded OOK in the 902–928 MHz band [3, 4].

Relationship to prior work. The most direct precedent is Rouf et al., “Neighborhood Watch” (ACM CCS 2012) [5]. That study reverse-engineered ERT-family AMR meters and found no confidentiality, integrity, or authentication

protections. It showed that an individual could monitor consumption from hundreds of homes with modest technical effort and identify unoccupied units in a multi-unit complex from broadcast data alone. Open-source decoders have since made reception routine: rtlamr [30] and rtl_433 [12] both decode ERT and Neptune R900 messages. This paper does not claim to discover that AMR telemetry is unencrypted.

Contributions. This paper contributes (1) a cross-sector taxonomy of 915 MHz cleartext protocols mapped to CISA's 16-sector framework, with each entry explicitly labeled as observed or documented and scored under CVSS v3.1; (2) field observations showing that current-generation formats (ERT SCMplus, Neptune R900 with E-Coder status fields) retain the same design assumptions as legacy ERT; (3) a precise account of which R900 status fields leak absence and leak-condition information in a single packet; and (4) an analysis of why these exposures fall outside the existing vulnerability-remediation and regulatory frameworks, illustrated by the “functions as designed” closure of VU#318962.

The remainder of this paper is organized as follows. Section II covers the regulatory framework and ERT architecture. Section III addresses ethics, legal scope, and data handling. Section IV describes methodology and capture coverage. Section V presents the device taxonomy. Section VI analyzes behavioral inference and scores each entry under CVSS v3.1. Section VII addresses remediation and the regulatory gap. Section VIII states limitations, and Section IX concludes.

II. Background and Regulatory Context

A. The 902–928 MHz ISM Band

The band is allocated under FCC Part 15, Subpart C, which imposes emission limits but no authentication or encryption requirements [1]. Part 15 devices must accept any interference received; this condition concerns radio-frequency interference with device operation and is not itself a legal basis for, or against, interception (see Section III). The band is shared by AMR meters, industrial telemetry radios, LoRaWAN, Z-Wave (908.42 and 916 MHz in North America, with Z-Wave Long Range at 912 and 920 MHz), and numerous proprietary protocols, using FHSS, DSSS, CSS, and narrowband FSK/OOK modulations. NIST SP 800-82 Rev. 3 notes that wireless networking in operational technology environments raises risk from adversaries in close physical proximity who lack direct access to equipment [8].

B. ERT Protocol Architecture

ERT uses OOK with Manchester encoding at a 32.768 kHz chip rate (approximately 16.4 kbps effective data rate), with no encryption or authentication beyond a 16-bit CRC [3, 4, 30]. The Standard Consumption Message (SCM) is 96 bits and carries a 26-bit endpoint ID, a 4-bit commodity type, two 2-bit tamper fields (physical and encoder), a 24-bit cumulative consumption value, and a 16-bit CRC. The Interval Data Message (IDM) is 92 bytes and adds 47 differential consumption intervals at 5-minute granularity, approximately four hours of history per packet [4, 30].

Endpoints in “bubble-up” mode transmit periodically without any interrogation signal [4]. This design supports drive-by reading and also allows any receiver in range to collect complete messages. Rouf et al. observed broadcasts approximately every 30 seconds from the meters they studied [5].

C. CISA Critical Infrastructure Sector Framework

PPD-21 (2013) established 16 critical infrastructure sectors. The April 2024 National Security Memorandum on Critical Infrastructure Security and Resilience superseded PPD-21, retained the 16-sector taxonomy, and designated CISA as National Coordinator; each sector has a Sector Risk Management Agency [9, 10]. The sectors are Chemical; Commercial Facilities; Communications; Critical Manufacturing; Dams; Defense Industrial Base; Emergency Services; Energy; Financial Services; Food and Agriculture; Government Facilities; Healthcare and Public Health; Information Technology; Nuclear Reactors, Materials, and Waste; Transportation Systems; and Water and Wastewater Systems. The author's captures directly observed exposure relevant to three sectors (Energy, Water and Wastewater, Food and Agriculture). Documented protocol exposure extends the relevant set to at least eight when adjacent-band vehicle telemetry is included (Section V).

III. Research Ethics, Legal Scope, and Data Handling

All observations were made with SDR hardware operated in receive-only mode. No transmissions, signal injection, replay, or interrogation were performed, and no utility infrastructure, customer account, or network resource was accessed or interfered with. Captures were made from the author's residence and publicly accessible locations.

Legal scope. The author sought independent legal counsel regarding this research. This paper makes no legal conclusion. Readers should note that U.S. federal law addresses radio reception in several places, including the Wiretap Act (18 U.S.C. § 2511), which contains exceptions for radio communications that are readily accessible to the general public, and 47 U.S.C. § 605, which restricts the divulgence and use of intercepted radio communications. How these provisions apply depends on the facts, including what is stored, retained, or disclosed. Nothing in this paper should be read as legal guidance.

Data handling. Published figures and datasets associated with this work are sanitized: endpoint identifiers are removed or pseudonymized, and no observation is linked to a street address or identifiable household. Aggregate statistics are reported in place of per-endpoint values wherever possible.

Vendor scope. Findings attributed to Neptune Technology Group, Itron Inc., and Landis+Gyr, the three vendors invited to the coordination case, include the author's own observations. Statements about all other manufacturers and protocols are drawn exclusively from public sources (manufacturer literature, FCC equipment authorizations, open-source decoder implementations, and published research) and are labeled *documented* in Tables I through III. They are not findings of this study.

IV. Passive Assessment Methodology

A. Hardware Platform

Captures used a generic RTL2832U-based SDR with a Rafael Micro R820T tuner, a commodity device typically priced between USD $25 and $60. The RTL2832U provides 8-bit samples and a tuning range of roughly 500 kHz to 1.766 GHz, with a stable instantaneous bandwidth of about 2.4 MHz. Captures in this study were centered at 915 MHz at 1 MSps. No license is required to receive ISM-band transmissions. Vaszary et al. (2021) similarly noted that SDR-based TPMS reception now costs under USD $50, a sharp reduction from equipment used in earlier research [11].

B. Software Toolchain

The primary decoder was rtl_433, an open-source multi-protocol ISM-band decoder supporting several hundred device protocols [12]. Relevant options include -vvv (verbose pulse timing), -F json (structured output), -M level (RSSI and SNR metadata), and -S unknown (saving unrecognized signals) [12, 13]. rtl_433 protocol numbers cited in Table I refer to the release used for this study and are recorded in the accompanying dataset documentation; protocol numbers can change between releases. Complementary tools include rtlamr (ERT SCM, SCM+, IDM, NetIDM, and R900) [30], Universal Radio Hacker for protocol reverse engineering [14], and gr-lora for identifying LoRa chirp spread spectrum signals even when payloads are encrypted [15].

C. Signal Fingerprinting for Undecoded Protocols

When rtl_433 cannot decode a transmission, its pulse analyzer (-A) characterizes pulse counts, widths, gap histograms, and frequency offsets; two distinct offsets suggest FSK while a single peak suggests OOK/ASK [13]. Useful discriminators include Z-Wave's fixed channel plan, LoRa's upward chirp preambles, and FHSS activity patterns. Presence detection without payload decoding still reveals device types at a site.

D. Capture Sessions and Coverage

Captures were collected in three sessions at a single residential location: 2026-02-20, 2026-02-24, and 2026-02-25 through 2026-02-26. Session durations and per-session counts are recorded in the dataset documentation. At 1 MSps the receiver observes roughly 1 MHz of the 26 MHz band at a time. Frequency-hopping endpoints such as the R900, which hops across 50 channels [27], are therefore received on only a small fraction of transmissions (on the order of a few percent, assuming uniform hopping). Even under this constraint, a single stationary receiver collected repeated

readings from multiple endpoints, which indicates that wider-band or multi-receiver collection would yield substantially more.

V. Device Taxonomy: CISA Sector Mapping

A. Energy Sector: Automatic Meter Reading Endpoints (observed)

Electric and gas meters are among the densest cleartext telemetry sources in the band. Itron reports a large North American installed base of ERT-compatible endpoints [16]. All ERT SCM, SCM+, and IDM transmissions are sent without cryptographic protection [3, 4, 30]. The author's captures decoded ERT-SCM (Type 12, gas), SCMplus, and Landis+Gyr GS-series frames. The presence of SCMplus, the successor to SCM, shows that the design assumptions persist in current-generation deployments and are not limited to legacy hardware.

Table I: AMR/Metering Protocols—Energy and Water Sector Exposure

Protocol	rtl_433 #	Modulation	Plaintext fields	CISA sector	Basis
ERT SCM	149	OOK / Manchester	26-bit endpoint ID, commodity type, 2×2-bit tamper, 24-bit consumption, CRC-16	Energy / Water	Observed
ERT SCM+	154	OOK / Manchester	32-bit endpoint ID, protocol ID, consumption, 16-bit tamper, CRC-CCITT	Energy / Water	Observed
ERT IDM	160	OOK / Manchester	32-bit serial, 47 × 5-min intervals (~4 h), tamper counters, outage flags	Energy	Documented
ERT NetIDM	161	OOK / Manchester	IDM fields plus consumption/generation intervals (reveals rooftop solar)	Energy	Documented
Neptune R900	228	FHSS, 50 ch. [27]	Endpoint ID, cumulative consumption, leak status (24 h), days of leak, days of no flow and reverse flow (35-day window)	Water / Wastewater	Observed
Landis+Gyr GS-series	271–273	FSK	Meter WAN address, meter ID, energy reading, subtype, CRC-16	Energy	Observed
Badger ORION	223	FSK (per decoder)	Meter ID, water consumption	Water / Wastewater	Documented
Mueller Hot Rod	255	GFSK (per decoder)	Meter ID, consumption, status flags	Water / Wastewater	Documented

Occupancy inference from smart meter data is well established. Kleiminger et al. (2013) detected occupancy from electricity consumption data [17]. Razavi et al. (2019) analyzed more than 5,000 households over 18 months and found high predictive power for both present and future occupancy [18]. Greveler et al. (2012) showed that at 0.5-second sampling, power profiles can reveal which television content a household is watching [19]; that resolution far exceeds anything broadcast by AMR endpoints and is cited here only to illustrate the upper bound of consumption-data inference. These studies used utility-side AMI data, not passively received AMR broadcasts. Their relevance is that ERT IDM packets carry 5-minute interval data comparable to or finer than the resolutions those studies used, in a single over-the-air message. Kalogridis et al. (2010) and McDaniel and McLaughlin (2009) set out the underlying threat model [20, 21].

B. Water and Wastewater Sector: Neptune R900 (observed)

The Neptune R900 is an FCC Part 15.247 frequency-hopping device that transmits across 50 channels in the 902–928 MHz band [27]. When paired with an E-Coder register, R900 messages carry the meter reading and status fields that Neptune describes in its own product literature [7, 27, 28]:

- **Leak status.** The E-Coder samples its register every 15 minutes. An intermittent leak is flagged when water was used in at least 50 of the 96 15-minute intervals in a 24-hour period; a continuous leak is flagged when water was used in all 96 [27]. The broadcast field is this 24-hour summary, not the underlying interval data.
- **Days of no flow and reverse flow.** These are reported over a 35-day window [28]. A current, non-zero consecutive-no-flow indication is available from a single received packet.

The discovery originated during reverse engineering of Fine Offset WH51 soil moisture sensors, when R900 transmissions from neighboring meters were decoded as an incidental observation. Across the three sessions, the author decoded transmissions from at least nine unique R900 endpoints. Most exhibited non-zero leak indicators

throughout the capture window. This is consistent with Neptune's own published deployment data: in one utility case study covering 901 accounts, fewer than half reported no leak condition of any kind [29]. The observation is therefore reported as corroborating that leak status is broadly visible to passive observers, not as a claim about the condition of any specific property.

Absence detection from water data does not require interval analysis in this case. The no-flow field summarizes recent absence directly, and the leak and flow indicators reveal household-level plumbing conditions, all in cleartext.

C. Energy, Water, and Chemical Sectors: 900 MHz Industrial Telemetry Radios (documented)

License-free 900 MHz radios from several manufacturers are marketed for pipeline, water treatment, and utility telemetry, typically carrying serial protocols such as Modbus RTU, DNP3, and DF1. These application protocols have no native encryption or authentication in their legacy forms. Many current radios offer link-layer encryption (commonly AES) and use proprietary frequency-hopping patterns, so exposure depends on configuration: where encryption is disabled or unsupported, process variables such as pressures, flow rates, valve positions, and alarm states can be exposed. The author did not observe industrial telemetry in this study. This entry is included because NIST SP 800-82 Rev. 3 identifies proximity-based wireless risk in OT environments [8] and because the same receive-only methodology applies. Its risk rating and CVSS score in Tables II and III assume encryption is not enabled.

D. Food and Agriculture Sector: Soil Moisture and Environmental Sensors (observed)

Fine Offset sensors, sold under Ecowitt, Ambient Weather, and other labels, are commonly encountered at 915 MHz. The WH51 soil moisture sensor broadcasts moisture percentage, raw ADC value, battery voltage, a boost flag, and a device ID in cleartext at roughly 70-second intervals [12]. That R900 transmissions surfaced during WH51 work illustrates that ISM-band assessment inherently reveals co-located infrastructure across sector boundaries.

E. Commercial and Government Facilities: Building Automation (documented)

Insteon is a dual-band home automation protocol operating at 915 MHz RF and over powerline. Its RF messages expose source and destination device addresses and command bytes in cleartext [12], from which lighting, thermostat, and motion events can be reconstructed. Insteon is primarily a residential and small-commercial product.

Z-Wave supports AES-128 under the S2 security framework, but legacy exposure remains. Under S0, the network key is exchanged encrypted with an all-zero key and can therefore be recovered by an observer present during pairing. Pen Test Partners (2018) showed that S2-capable devices can be downgraded to S0 during pairing (“Z-Shave”) [23]. CERT/CC VU#142629 (2022) documents further issues in Silicon Labs Z-Wave chipsets, including 100/200/300-series chipsets that do not support encryption at all [24]. Pen Test Partners cited more than 100 million Z-Wave chips in the field at the time of their research [23].

LoRaWAN encrypts application payloads with AES-128, but the DevAddr is visible in every uplink and the DevEUI is visible in over-the-air join requests, enabling device enumeration and traffic analysis.

F. Adjacent Band: Transportation Systems Sector—TPMS (documented)

Tire Pressure Monitoring Systems operate at 315 MHz in the United States and 433 MHz in Europe, outside the 915 MHz band that is the focus of this paper; they are included because the same hardware and methodology apply with a frequency change. TPMS became mandatory in the U.S. under the TREAD Act, with FMVSS No. 138 requirements fully phased in for model-year 2008 vehicles. Rouf et al. (USENIX Security 2010) found that TPMS messages carry static 32-bit identifiers, can be received at roughly 40 meters from a passing vehicle, can be triggered remotely, and lack authentication; they demonstrated spoofed pressure warnings in a moving vehicle [22]. Vaszary et al. (2021) reproduced ID tracking at up to 40 meters using a low-cost SDR [11].

VI. Behavioral Inference Capabilities

A. Occupancy Detection from Utility Meter Data

Farokhi (2020) reviews the smart meter privacy literature and confirms that simple analysis of consumption patterns can reveal household occupancy, with higher-resolution data enabling appliance-level disaggregation through non-intrusive load monitoring [25]. McKenna et al. (2012) similarly identify consumption data as a strong indicator of

occupancy [26]. Rouf et al. demonstrated the attack against AMR broadcasts specifically, identifying unoccupied residences from passively received data [5].

The concern scales with ERT IDM data, which delivers 47 intervals at 5-minute resolution per packet. Continuous collection across a service territory could, in principle, yield near-real-time occupancy estimates for many residences. For the R900, the relevant exposure is more direct: the no-flow and leak status fields summarize absence and household water use without any aggregation by the observer.

B. Vehicle Presence and Pattern-of-Life Tracking (adjacent band)

Because TPMS identifiers are static and each vehicle carries four sensors, a set of fixed receivers at chokepoints such as entry roads or parking structures can associate recurring identifier sets with individual vehicles [11, 22]. Unlike optical license plate recognition, this does not require line of sight and is not affected by darkness or obscured plates. Identifiers typically persist until sensors are replaced.

C. Risk Prioritization Matrix

Table II: 915 MHz Passive Intelligence Risk Prioritization—CISA Sector Mapping

Device / Protocol	CISA sector	Encryption	Data exposed	Inference capability	Risk	CVSS	Basis
ERT IDM (electric)	Energy	None	47 × 5-min intervals, outage flags, tamper	Occupancy, appliance use, solar	Critical	6.5	Documented
Neptune R900 + E-Coder	Water / WW	None	Consumption, leak status, no-flow and reverse-flow days	Absence, household water use, plumbing faults	High	6.5	Observed
900 MHz telemetry radios (encryption off)	Energy / Water / Chemical / Dams	Optional; config-dependent	Modbus/DNP3 process values, alarms	Industrial process profiling	High*	5.3	Documented
ERT SCM / SCM+	Energy	None	Cumulative consumption, ID, tamper	Usage trending, site profiling	High	4.3	Observed
Landis+Gyr GS-series	Energy	None	WAN address, meter ID, energy reading	Network segment mapping, consumption trending	High	4.3	Observed
ERT NetIDM	Energy	None	Consumption and generation intervals	Distributed solar profiling	High	6.5	Documented
TPMS (315/433 MHz)	Transportation	None	Static 32-bit sensor ID, pressure, temp	Vehicle tracking, pattern of life	High	4.3	Documented
Insteon (915 MHz)	Commercial Facilities	None	Device addresses, commands	Occupancy, routines	Medium	4.3	Documented
Z-Wave (S0)	Commercial / Government Facilities	S2 AES-128; S0 weak; legacy none	Presence; S0 traffic decryptable after pairing capture	Automation topology, access events	Medium	5.3	Documented
LoRaWAN	Water / Ag / Energy	AES-128 payload	DevAddr; DevEUI in join requests	Device enumeration, traffic analysis	Medium	4.3	Documented
Fine Offset WH51	Food & Agriculture	None	Soil moisture, device ID, battery	Irrigation schedule	Low–Med	4.3	Observed
DSC PowerG	Commercial / Government Facilities	AES-128 + FHSS	RF presence only	Security system presence	Low	0.0	Documented

Ratings are the author's qualitative assessment of passive collection risk. They are not official CISA, government, or vendor classifications. “Observed” means decoded in the author's receive-only captures; “Documented” means derived solely from public sources. *Rating assumes link encryption is not enabled; correctly configured encrypted links reduce exposure substantially. CVSS 3.1 is the author-assessed base score; vectors and rationale are in Table III.

D. CVSS v3.1 Scoring

To support comparison with conventional vulnerability tracking, each entry in Table II was also scored under the Common Vulnerability Scoring System v3.1 [31]. No CVE was assigned in VU#318962, so these scores are the author's assessment. They have not been assigned or reviewed by a CVE Numbering Authority, NVD, CERT/CC, or

the affected vendors.

Scoring assumptions. Scores follow the receive-only threat model used throughout this paper. Attack Vector is Adjacent (AV:A), since reception requires RF proximity; the specification treats wireless links this way [31]. Privileges Required and User Interaction are None, and Scope is Unchanged. Attack Complexity is Low where a commodity SDR and an open-source decoder suffice, and High where the observer must first characterize a proprietary hopping pattern or be present during device pairing. Integrity and Availability are None because the observer never transmits. Confidentiality is High where a single message discloses information with a direct, serious impact, such as interval consumption or absence indicators, and Low where only cumulative readings, identifiers, or sensor values are exposed.

Table III: CVSS v3.1 Scores for Passive Collection (Author-Assessed)

Device / Protocol	Vector (CVSS:3.1/…)	Base	Env. (CR:H)	Rationale
ERT IDM (electric)	AV:A/AC:L/PR:N/UI:N/S:U/C:H/I:N/A:N	6.5 Medium	8.3 High	One packet carries ~4 h of 5-min interval data supporting occupancy and appliance inference (C:H).
Neptune R900 + E-Coder	AV:A/AC:L/PR:N/UI:N/S:U/C:H/I:N/A:N	6.5 Medium	8.3 High	No-flow days directly disclose absence; leak status discloses plumbing condition (C:H).
900 MHz telemetry radios (encryption off)	AV:A/AC:H/PR:N/UI:N/S:U/C:H/I:N/A:N	5.3 Medium	7.1 High	Process values and alarms exposed (C:H); proprietary hopping must first be characterized (AC:H).
ERT SCM / SCM+	AV:A/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N	4.3 Medium	5.0 Medium	Cumulative reading, ID, and tamper flags only; trending requires repeated collection (C:L).
Landis+Gyr GS-series	AV:A/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N	4.3 Medium	5.0 Medium	Identifiers and cumulative energy reading (C:L).
ERT NetIDM	AV:A/AC:L/PR:N/UI:N/S:U/C:H/I:N/A:N	6.5 Medium	8.3 High	Interval consumption and generation data in a single message (C:H).
TPMS (315/433 MHz)	AV:A/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N	4.3 Medium	5.0 Medium	Static sensor ID and readings (C:L); tracking requires multi-receiver collection, which CVSS does not score.
Insteon (915 MHz)	AV:A/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N	4.3 Medium	5.0 Medium	Device addresses and command bytes (C:L).
Z-Wave (S0)	AV:A/AC:H/PR:N/UI:N/S:U/C:H/I:N/A:N	5.3 Medium	7.1 High	Network key recoverable only if present during pairing (AC:H); traffic then decryptable (C:H).
LoRaWAN	AV:A/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N	4.3 Medium	5.0 Medium	Payload encrypted; DevAddr and DevEUI exposed (C:L).
Fine Offset WH51	AV:A/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N	4.3 Medium	5.0 Medium	Soil moisture, battery, and device ID (C:L).
DSC PowerG	AV:A/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:N	0.0 None	0.0 None	Encrypted; RF presence alone is not a confidentiality impact (C:N).

All vectors carry the prefix CVSS:3.1/. Env. (CR:H) is the environmental score with Confidentiality Requirement set to High and all other metrics unmodified. Scores computed with the FIRST v3.1 equations [31]. They describe receive-only exposure; active attacks are not scored.

What the base score misses. A passive, adjacent, confidentiality-only exposure cannot score above 6.5 under CVSS v3.1. The two entries rated highest in Table II, ERT IDM (Critical) and the Neptune R900 (High), both sit at that ceiling and are classed Medium. The gap is structural. CVSS base metrics score one vulnerable component at one point in time. They do not capture aggregation across a service territory, persistence over the service life of hardware that cannot be patched, or the inference capability described in Section VI-A. The qualitative ratings in Table II include those factors. The base scores do not.

Temporal and environmental context. Temporal metrics do not lower the observed entries. Functional open-source decoders are public (E:H) [12, 30], no fix is offered (RL:U), and the behavior is confirmed by vendor literature (RC:C), so temporal scores equal base scores. Environmental metrics are the intended mechanism for deployment context. With Confidentiality Requirement set to High, appropriate for endpoints near government facilities or other sensitive sites, the C:H entries rise to 8.3 (7.1 where Attack Complexity is High) and the C:L entries to 5.0. Utilities

and asset owners should score these exposures in their own environments rather than rely on base scores.

Active attacks are out of scope. Rouf et al. showed that AMR packets can be spoofed [5] and demonstrated spoofed TPMS warnings [22], and unauthenticated Modbus or DNP3 links accept injected commands. Scoring those would add Integrity impact, and for industrial links Availability impact; adding I:L to the ERT vector alone yields 5.4. These vectors are noted for completeness. They were not tested.

VII. Remediation and the Regulatory Gap

A. Technical Constraints on Remediation

For deployed AMR endpoints the constraint is architectural. Many are battery-powered with limited computation and no field firmware update path, so the broadcast format cannot be changed without replacing hardware. This stands in contrast to software vulnerabilities that vendors can patch remotely. Remediation for AMR therefore means utility-funded endpoint replacement, at a large aggregate cost across the industry.

Migration from one-way AMR to two-way AMI is the recognized path; AMI deployments using protocols such as ANSI C12.22 or DLMS/COSEM with cryptographic protection address cleartext transmission. However, utilities have limited economic incentive to replace functioning meters early absent a mandate, and some AMI migration paths are designed to retain compatibility with existing AMR endpoints.

B. The Regulatory Gap

No current U.S. federal regulation requires cryptographic protection for 902–928 MHz AMR transmissions. FCC Part 15 imposes no encryption requirement. NERC CIP applies to bulk electric system assets, not customer premises metering. EPA drinking water regulations contain no provisions for wireless telemetry security. NIST SP 800-82 Rev. 3 is advisory [8]. The 2024 National Security Memorandum calls for elevating minimum security and resilience requirements across critical infrastructure [9], but no implementing requirement for AMR encryption has followed.

The disposition of VU#318962 illustrates the gap. Because the affected products behave as designed, the coordinated disclosure process had no defect to remediate and closed the case without an advisory. When the insecurity is the design, the vulnerability pipeline cannot resolve it; only standards or regulation can. The CVSS results in Section VI-D point the same way: a scoring system built to rank defects reports a design-level, territory-wide exposure as Medium.

C. Interim Mitigations

- Inventory ERT/AMR endpoint deployment in service territories that include sensitive sites (government facilities, critical infrastructure), and prioritize those areas for AMI migration.
- Where supported, configure endpoints for wake-up mode rather than bubble-up mode to reduce continuous broadcast exposure, recognizing that transmissions during interrogation remain receivable.
- Recognize that passive collection is generally not detectable. Mitigation must focus on reducing emissions and the sensitivity of what is emitted, not on detecting observers.
- For 900 MHz industrial telemetry links, enable link-layer encryption with authentication where supported, or add application-layer protection (for example, an IPsec overlay); evaluate wired alternatives for the most critical control links.
- For Z-Wave, require S2, replace S0-only and non-encrypting legacy devices, and restrict pairing to supervised maintenance windows.

VIII. Limitations

Field observations come from a single residential location over three sessions with a single narrowband receiver, and cover a small fraction of frequency-hopping transmissions. Endpoint counts are lower bounds. Decoded field values were interpreted using open-source decoders and vendor literature and were not validated against utility records. Industrial telemetry radios, building automation systems, LoRaWAN, and TPMS were not observed; their entries rely on public sources. Occupancy-inference results cited here were obtained on utility-side AMI datasets, and their accuracy on passively received AMR data has not been measured in this work. Risk ratings are qualitative. CVSS

scores are the author's assessment; reasonable analysts may differ, particularly on the Confidentiality impact, and the scores cover receive-only exposure only.

IX. Conclusion

The 902–928 MHz ISM band exposes cleartext telemetry of real consequence to anyone with a receiver costing less than USD $60. This study directly observed utility meter and agricultural sensor telemetry in the Energy, Water and Wastewater, and Food and Agriculture sectors. Public documentation indicates comparable exposure in building automation and, where encryption is not enabled, in industrial telemetry links, with similar issues in adjacent-band vehicle systems.

The Neptune R900 findings in VU#318962 are not an anomaly but a representative instance of an architectural choice made before commodity SDR receivers existed. The research community has documented the privacy implications of cleartext AMR since at least 2010 [5, 18, 20, 22], and the coordination process has confirmed that these products function as designed. That is precisely why the remedy must be policy rather than patching. We call on CISA, EPA, NERC, and the FCC to consider minimum encryption and authentication standards for critical infrastructure telemetry operating in unlicensed ISM bands, beginning with new procurement and risk-prioritized replacement near sensitive sites.

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